

Purpose

This document defines the synchronization model governing the exchange of information between Kommo CRM and Aloware.

Its purpose is to establish how data moves across the integration, which system owns each business entity, when synchronization occurs, and the principles that ensure consistency between platforms.

This document describes synchronization behavior at a conceptual level and intentionally excludes implementation details such as field mappings, API endpoints, validation logic, and workflow configuration.

Scope

This document defines:

- Synchronization directions
- Data ownership
- Synchronization triggers
- Synchronization frequency
- Consistency model
- Conflict resolution principles
- Synchronization boundaries

Implementation details are documented separately.

Synchronization Objectives

The synchronization model has been designed to achieve the following objectives:

- Maintain a single source of truth for every business entity.
- Keep both platforms operationally synchronized.
- Prevent ownership conflicts.

- Minimize duplicate information.
- Preserve data integrity.
- Support scalable event-driven integrations.
- Reduce unnecessary API operations.

Synchronization Strategy

The integration follows a **bidirectional synchronization model**.

Although information travels in both directions, ownership remains clearly separated.

Each platform is responsible only for the business information belonging to its domain.

Synchronization is therefore based on **data ownership**, not on full data replication.

Data Ownership Model

The following ownership model governs every synchronization process.

Business Domain	System of Record
Leads	Kommo CRM
Contacts	Kommo CRM
Companies	Kommo CRM
Sales Pipeline	Kommo CRM
Lead Status	Kommo CRM
Sales Owner	Kommo CRM
Calls	Aloware
SMS	Aloware
Call Recordings	Aloware
AI Call Summary	Aloware
Communication History	Aloware

Ownership is exclusive.

Information may be synchronized but ownership never changes.

Synchronization Directions

Kommo → Aloware

Purpose:

Synchronize commercial information required by communication workflows.

Typical synchronized information includes:

- Lead lifecycle
- Contact information
- Pipeline stage
- Owner assignment
- Business attributes

Kommo remains the authoritative source.

Aloware → Kommo

Purpose:

Synchronize communication results generated by customer interactions.

Typical synchronized information includes:

- Call activity
- SMS activity
- Call recordings
- AI-generated summaries
- Communication outcomes

Aloware remains the authoritative source.

Synchronization Triggers

Synchronization is initiated by business events.

Preferred trigger mechanism:

- Webhooks

Fallback mechanism:

- Scheduled polling when event notifications are unavailable.

Every synchronization begins only after a relevant business event occurs.

Synchronization Frequency

The solution is designed for near real-time synchronization.

Characteristics include:

- Event-driven execution
- Independent processing
- No scheduled full synchronization
- No continuous replication

Small processing delays are considered acceptable.

Consistency Model

The integration follows an **Eventual Consistency** model.

Updates are propagated after business events are processed successfully.

Temporary differences between systems are acceptable while synchronization is in progress.

Consistency is restored automatically after successful processing.

Synchronization Boundaries

Synchronization only exchanges information required by the integration objectives.

The solution does not attempt to mirror entire platform datasets.

Only business entities participating in defined integration flows are synchronized.

Conflict Resolution Principles

The synchronization model minimizes conflicts through clear ownership rules.

General principles include:

- The System of Record always has priority.
- Non-owning systems must not overwrite authoritative information.
- Communication data never replaces commercial data.
- Commercial data never replaces communication history.
- Independent domains remain isolated.

Ownership determines update authority.

Update Strategy

Synchronization follows an incremental update model.

Characteristics include:

- Event-based updates
- Partial synchronization
- Entity-level processing
- No full database replication
- No bulk overwrite operations

Only information affected by the triggering event is synchronized.

Duplicate Prevention

The synchronization model assumes that duplicate processing may occur.

To minimize duplicate business effects:

- Events should be processed idempotently.
- Existing records should be identified before creation.
- Synchronization should prioritize updates over unnecessary record creation.

Implementation details are defined separately.

Failure Behavior

Synchronization failures do not invalidate ownership.

If synchronization cannot be completed:

- The source system remains authoritative.

- The destination system may become temporarily outdated.
- Consistency is restored after successful reprocessing.

Failures affect only the current synchronization event.

Scalability

The synchronization model supports future expansion.

Additional synchronization domains may be incorporated without modifying the ownership principles.

Examples include:

- Appointments
- Marketing activities
- Additional AI events
- Customer engagement metrics
- Future communication channels

All future integrations should follow the same synchronization model.

Design Principles

The synchronization model is based on the following principles:

- Single Source of Truth
 - Clear Ownership
 - Event-Driven Synchronization
 - Eventual Consistency
 - Loose Coupling
 - Independent Processing
 - Incremental Updates
 - Idempotent Operations
 - Failure Isolation
 - Scalability
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Out of Scope

The following topics are intentionally excluded:

- Field mappings
- Stage mappings
- Agent mappings
- Data transformation logic
- Validation rules
- Retry implementation
- Error handling
- Logging
- Monitoring
- API implementation
- Make scenario configuration

Related Documents

This document should be read together with:

- Architecture Design
- Integration Flows Design
- Field Mapping
- Stage Mapping
- Agent Mapping
- Data Transformation Rules
- Validation Rules
- Error Handling Strategy
- Retry Strategy
- Logging Strategy
- Monitoring Strategy
- Security Considerations

Field Mapping

Version: 1.0

Status: Draft

Phase: 2 — Technical Design

Purpose

This document defines how business information is mapped between Kommo CRM and Aloware.

Its purpose is to identify which business entities and attributes participate in the integration, establish their conceptual correspondence between platforms, and define the ownership of each mapped field.

This document focuses on **what is mapped**, not **how it is transformed**. Data conversion, formatting, and normalization rules are documented separately.

Scope

This document defines:

- Business entities participating in synchronization
- Conceptual field correspondence
- Field ownership
- Synchronization direction
- Mapping principles

This document intentionally excludes:

- Data transformations
 - Validation rules
 - API payloads
 - Endpoint specifications
 - Implementation details
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Mapping Objectives

The field mapping model has the following objectives:

- Maintain semantic consistency between platforms.

- Clearly identify the origin of every synchronized value.
 - Avoid duplicate ownership.
 - Simplify future maintenance.
 - Facilitate implementation and testing.
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Mapping Principles

Conceptual Mapping

Mappings are based on business meaning rather than field names.

Equivalent information may have different representations in each platform.

Ownership Preservation

Each mapped field has a single authoritative source.

The destination system must never overwrite information owned by the source system.

Directional Mapping

Mappings are directional.

A field may be synchronized:

- Kommo → Aloware
 - Aloware → Kommo
 - Both directions (only when ownership remains unambiguous)
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Selective Synchronization

Only fields required by the integration are synchronized.

Internal platform attributes remain outside the integration scope.

Business Entity Mapping

The integration currently involves the following business entities.

Business Entity	Kommo	Aloware	Synchronization
Lead	✓	Reference	Kommo → Aloware
Contact	✓	✓	Kommo → Aloware
Company	✓	Reference	Kommo → Aloware
Sales Owner	✓	Agent	Kommo → Aloware
Pipeline Stage	✓	List / Campaign	Kommo → Aloware
Call Activity	Reference	✓	Aloware → Kommo
SMS Activity	Reference	✓	Aloware → Kommo
Call Recording	Reference	✓	Aloware → Kommo
AI Summary	Reference	✓	Aloware → Kommo

Conceptual Field Categories

Instead of defining individual fields at this stage, mappings are organized into logical categories.

Identification Fields

Examples include:

- Lead ID
- Contact ID
- Company ID
- External IDs
- Phone identifiers

These fields uniquely identify synchronized entities.

Contact Information

Examples include:

- Name

- Phone number
- Email
- Company association

Used to identify communication recipients.

Commercial Information

Examples include:

- Pipeline
- Stage
- Sales owner
- Lead status
- Commercial attributes

Owned by Kommo.

Communication Information

Examples include:

- Call result
- Call duration
- Recording URL
- SMS content
- AI Summary

Owned by Aloware.

Metadata

Examples include:

- Creation timestamps
- Update timestamps
- Source platform
- Synchronization identifiers

Used for operational purposes.

Mapping Matrix

The following matrix defines the conceptual ownership of synchronized information.

Field Category	Owner	Direction
Lead Information	Kommo	Kommo → Aloware
Contact Information	Kommo	Kommo → Aloware
Company Information	Kommo	Kommo → Aloware
Sales Assignment	Kommo	Kommo → Aloware
Pipeline Information	Kommo	Kommo → Aloware
Call Data	Aloware	Aloware → Kommo
SMS Data	Aloware	Aloware → Kommo
Recording Data	Aloware	Aloware → Kommo
AI Summary	Aloware	Aloware → Kommo

Mapping Characteristics

The field mapping follows these characteristics.

Explicit

Every synchronized field must have a documented correspondence.

Traceable

The origin of every synchronized value must be identifiable.

Deterministic

A field always maps to the same destination attribute.

Maintainable

Changes to mappings should affect only the corresponding mapping definition.

Extensible

Additional fields may be incorporated without redesigning the mapping model.

Mapping Constraints

The following constraints apply.

- Fields without business value should not be synchronized.
 - Computed values should remain in their originating system unless explicitly required.
 - Internal platform identifiers should not be exposed unnecessarily.
 - Mapping definitions should remain independent of API implementation details.
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Future Detailed Mapping

Detailed attribute-level mappings will be documented during implementation.

Typical information includes:

- Source field
- Destination field
- Data type
- Required status
- Default value
- Nullable behavior
- Synchronization direction
- Transformation reference

This document serves as the conceptual foundation for those implementation mappings.

Related Documents

This document should be read together with:

- Synchronization Model
- Stage Mapping
- Agent Mapping
- Data Transformation Rules
- Validation Rules
- API Objects
- Data Mapping Requirements

Stage Mapping

Version: 1.0

Status: Draft

Phase: 2 — Technical Design

Purpose

This document defines the conceptual mapping between the sales pipeline stages managed in Kommo CRM and the operational states, lists, or communication workflows managed in Aloware.

Its purpose is to establish how changes in the commercial lifecycle are translated into communication actions while preserving the ownership of each platform.

This document defines **which business states correspond between systems**, not the automation logic used to process them.

Scope

This document defines:

- Stage ownership
- Stage synchronization principles
- Conceptual stage correspondence
- Synchronization direction
- Mapping constraints

This document intentionally excludes:

- Make scenarios
 - API implementation
 - Workflow configuration
 - Automation rules
 - Validation logic
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Objectives

The stage mapping model has the following objectives:

- Maintain alignment between sales and communication processes.

- Ensure that communication workflows reflect the current commercial status.
- Avoid ambiguous stage relationships.
- Preserve a single commercial source of truth.
- Simplify future pipeline changes.

Ownership

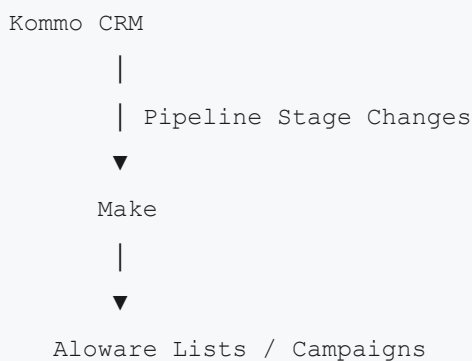
Pipeline stages belong exclusively to Kommo CRM.

Aloware consumes this information to organize communication activities but does not own or manage the commercial lifecycle.

Domain	System of Record
Sales Pipeline	Kommo CRM
Lead Stage	Kommo CRM
Communication Lists	Aloware
Dialing Campaigns	Aloware

Stage synchronization is therefore **unidirectional**.

Synchronization Direction



Changes originating in Aloware must never modify the sales pipeline in Kommo.

Mapping Strategy

The integration follows a **business-state mapping** strategy.

Rather than attempting to replicate the pipeline structure, each commercial stage is associated with the operational communication state that best represents the desired customer interaction.

This approach allows both systems to evolve independently while maintaining consistent behavior.

Conceptual Mapping

Each relevant sales stage may correspond to one operational destination in Aloware.

Kommo Business Stage	Aloware Operational State
New Lead	Initial Contact Queue
Contact Attempt	Outbound Calling Queue
Qualified Lead	Active Follow-up Queue
Proposal Sent	Follow-up Campaign
Negotiation	Priority Contact Queue
Won	Customer Communication Flow
Lost	Closed / Archived

The exact stage names will be documented in the implementation specification.

Mapping Characteristics

One-Way Synchronization

Commercial stages are synchronized only from Kommo to Aloware.

Explicit Mapping

Every synchronized stage must have an explicitly defined destination.

Undefined mappings should never rely on implicit behavior.

Stable Relationships

Each business stage should always produce the same operational destination.

Mappings should remain predictable and deterministic.

Business-Oriented

Mappings represent business intent rather than technical implementation.

Operational states should reflect the communication strategy associated with each sales stage.

Synchronization Behavior

When a mapped stage changes in Kommo:

1. The business event is generated.
2. Make receives the event.
3. The stage mapping is evaluated.
4. The corresponding operational destination is identified.
5. Aloware is updated accordingly.

Only the affected lead is synchronized.

Unmapped Stages

Not every stage necessarily participates in the integration.

Stages without communication relevance may be excluded.

If a stage has no defined mapping:

- No synchronization is executed.
- The event is ignored.
- No communication workflow is modified.

This prevents unintended automation.

Pipeline Evolution

Sales pipelines may evolve over time.

To support future changes:

- Stage mappings should remain independent of implementation.
 - New stages should be documented before activation.
 - Removed stages should be deprecated in the mapping documentation.
 - Existing mappings should remain stable whenever possible.
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Design Principles

The stage mapping model follows these principles:

- Kommo owns the commercial lifecycle.
 - Aloware reflects communication readiness.
 - One business stage produces one predictable operational outcome.
 - Mappings are explicit and documented.
 - Business meaning takes precedence over technical naming.
 - Pipeline evolution should not require architectural redesign.
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Future Detailed Mapping

The implementation phase will introduce a detailed mapping specification including:

- Kommo Pipeline ID
- Kommo Stage ID
- Kommo Stage Name
- Aloware List ID
- Aloware Campaign ID
- Synchronization Trigger
- Processing Conditions
- Notes

This document serves as the conceptual foundation for that specification.

Related Documents

This document should be read together with:

- Synchronization Model
- Field Mapping
- Agent Mapping

- Integration Flows Design
- Data Transformation Rules

Agent Mapping

Version: 1.0

Status: Draft

Phase: 2 — Technical Design

Purpose

This document defines the conceptual mapping between users in Kommo CRM and agents in Aloware.

Its purpose is to establish how ownership and responsibility for customer interactions are synchronized across both platforms while preserving a single authoritative source for user assignments.

This document defines **who corresponds to whom**, not the implementation details of user synchronization.

Scope

This document defines:

- User ownership
- Agent correspondence
- Assignment synchronization
- Mapping principles
- Mapping constraints

This document intentionally excludes:

- User provisioning
 - Authentication
 - Permission management
 - API implementation
 - Workflow configuration
 - Synchronization logic
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Objectives

The agent mapping model has the following objectives:

- Maintain consistent ownership across platforms.

- Ensure communication activities are assigned to the correct agent.
- Prevent assignment conflicts.
- Support scalable user management.
- Simplify future organizational changes.

Ownership

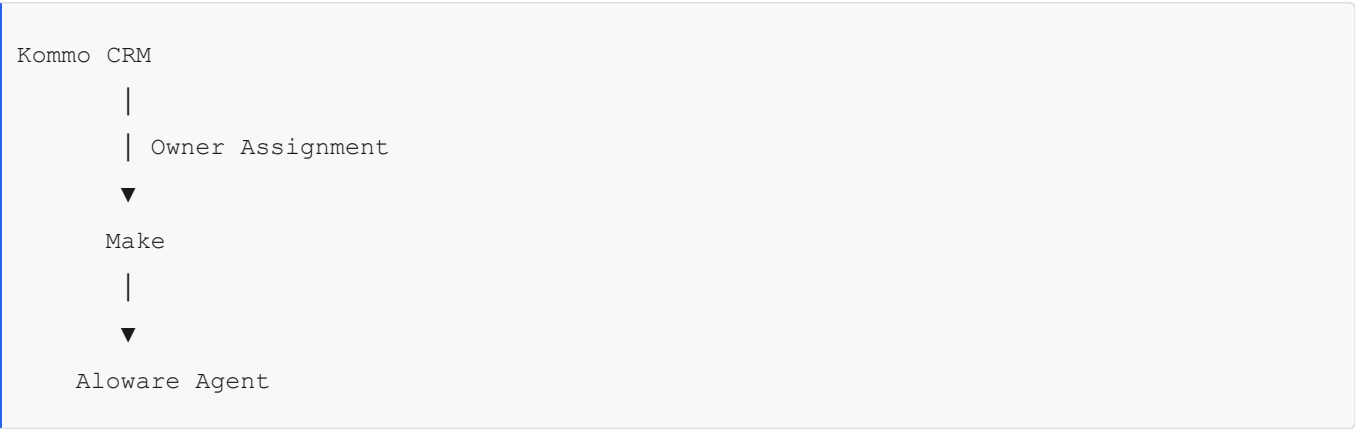
User assignments are owned by Kommo CRM.

Aloware receives agent assignments to execute communication activities but does not determine commercial ownership.

Domain	System of Record
Sales Owner	Kommo CRM
Lead Assignment	Kommo CRM
Communication Agent	Aloware

Ownership is never transferred.

Synchronization Direction



Assignments originate in Kommo and are propagated to Aloware.

Changes performed directly in Aloware should not modify ownership in Kommo.

Mapping Strategy

The integration follows a **one-to-one conceptual mapping** between sales owners and communication agents whenever possible.

Each commercial owner is associated with a corresponding operational agent responsible for customer communications.

This mapping allows both systems to remain aligned while preserving their independent user management.

Conceptual Mapping

The mapping relates business responsibilities rather than platform-specific identifiers.

Kommo Role	Aloware Role
Sales Owner	Communication Agent
Account Manager	Assigned Agent
Sales Representative	Calling Agent
Team Member	Operational Agent

The exact user identifiers are documented separately during implementation.

Assignment Behavior

When the owner of a lead changes in Kommo:

1. The assignment event is generated.
2. Make receives the event.
3. The corresponding Aloware agent is identified.
4. The communication assignment is updated.
5. Future communication activities are associated with the mapped agent.

Only the affected lead is updated.

Mapping Characteristics

One-Way Synchronization

Assignments are synchronized exclusively from Kommo to Aloware.

Explicit Relationships

Every synchronized owner should have an explicitly defined corresponding agent.

Implicit mappings should be avoided.

Stable Assignments

A user should always map to the same operational agent unless the mapping configuration is intentionally updated.

Independent User Management

Each platform maintains its own user accounts.

The integration synchronizes assignments rather than user administration.

Unmapped Users

Some users may not participate in the integration.

Examples include:

- Administrators
- Read-only users
- Support personnel
- Inactive accounts

If no corresponding agent exists:

- The synchronization should not assign the lead automatically.
- The event should be logged for operational review.

- Existing assignments should remain unchanged unless otherwise defined.
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Organizational Changes

The mapping model supports organizational evolution.

Examples include:

- New sales representatives
- New communication agents
- Team restructuring
- User deactivation
- Role reassignment

Mapping definitions should be updated before organizational changes become operational.

Design Principles

The agent mapping model follows these principles:

- Kommo owns commercial assignments.
 - Aloware reflects operational responsibility.
 - One owner corresponds to one operational agent whenever possible.
 - User administration remains independent in each platform.
 - Mapping relationships are explicit and documented.
 - Assignment synchronization should be deterministic.
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Future Detailed Mapping

The implementation phase will introduce a detailed mapping specification including:

- Kommo User ID
- Kommo User Name
- Kommo Email
- Aloware Agent ID
- Aloware Agent Name
- Synchronization Status
- Active Flag
- Notes

This document serves as the conceptual foundation for that specification.

Related Documents

This document should be read together with:

- Synchronization Model
- Field Mapping
- Stage Mapping
- Data Transformation Rules
- Validation Rules
- Integration Flows Design

Purpose

This document defines the conceptual rules for transforming business data exchanged between Kommo CRM and Aloware.

Its purpose is to ensure that information maintains its business meaning when moving between platforms, regardless of differences in data models, formats, or terminology.

This document specifies **how data should be conceptually transformed**, not the technical implementation of those transformations.

Scope

This document defines:

- Transformation principles
- Data normalization rules
- Data conversion categories
- Enrichment guidelines
- Transformation constraints

This document intentionally excludes:

- Field mappings
 - Validation rules
 - API payloads
 - Platform-specific expressions
 - Make formulas
 - Implementation details
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Objectives

The transformation model has the following objectives:

- Preserve business meaning.
 - Normalize information across systems.
 - Reduce platform-specific dependencies.
 - Improve data consistency.
 - Support maintainable integrations.
 - Facilitate future platform evolution.
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Transformation Principles

Business Meaning Preservation

Every transformation must preserve the original business intent of the information.

Technical differences between platforms should never modify the meaning of the data.

Deterministic Transformation

The same input should always produce the same output.

Transformation rules must remain predictable and repeatable.

Platform Independence

Transformation logic should be defined independently of the implementation platform.

Business rules must remain valid whether the integration is implemented in Mule, n8n, or another orchestration platform.

Minimal Transformation

Only the transformations required to achieve compatibility between systems should be performed.

Unnecessary manipulation of data should be avoided.

Transformation Categories

Format Transformation

Converts data into the format expected by the destination platform.

Examples include:

- Date formats
 - Phone formats
 - Boolean representations
 - Number formatting
 - Text encoding
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Structural Transformation

Adapts differences in data structure.

Examples include:

- Object restructuring
 - Nested attributes
 - Collection mapping
 - Metadata organization
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Value Transformation

Converts business values between equivalent representations.

Examples include:

- Status values
 - Stage names
 - Communication results
 - Enumerations
 - Categories
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Identifier Transformation

Converts or associates identifiers required by each platform.

Examples include:

- External IDs
- Platform-specific IDs
- Owner references
- Campaign references

Identifier relationships must remain traceable throughout the synchronization lifecycle.

Metadata Transformation

Operational metadata may be generated or adapted to support synchronization.

Examples include:

- Synchronization timestamps
- Source platform indicators
- Correlation identifiers
- Processing metadata

Metadata should never alter business information.

Normalization Rules

Before data is synchronized, it should be normalized whenever appropriate.

Typical normalization includes:

- Standardizing phone numbers.
- Trimming unnecessary whitespace.
- Normalizing text encoding.
- Applying consistent date formats.
- Removing unsupported characters.
- Standardizing null or empty values.

Normalization improves consistency across systems.

Data Enrichment

The integration may enrich synchronized information when required.

Examples include:

- Resolving owner references.
- Resolving stage mappings.
- Resolving agent mappings.
- Adding synchronization metadata.
- Associating external identifiers.

Enrichment should only supplement existing information and must not modify business ownership.

Data Reduction

Not every source attribute must be transferred.

The integration may intentionally omit:

- Internal platform metadata.
- Temporary processing values.
- System-generated attributes.
- Unused configuration fields.
- Information outside the integration scope.

Only business-relevant information should be synchronized.

Null Handling

Transformation rules should define consistent handling of missing information.

General principles include:

- Preserve intentional null values.
- Avoid replacing missing values with arbitrary defaults.
- Ignore optional attributes when appropriate.
- Distinguish between "missing" and "empty" values.

The exact behavior is defined during implementation.

Transformation Constraints

The following constraints apply to every transformation.

- Transformations must not change data ownership.
 - Business meaning must remain unchanged.
 - Transformations should be reversible whenever practical.
 - Platform-specific logic should remain isolated.
 - Transformations must be deterministic.
 - Unsupported values should never produce ambiguous results.
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Design Principles

The transformation model follows these principles:

- Consistency
 - Simplicity
 - Predictability
 - Traceability
 - Platform Independence
 - Reusability
 - Maintainability
 - Scalability
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Future Detailed Transformation Specification

The implementation phase will define detailed transformation rules including:

- Source Field
- Destination Field
- Source Data Type
- Destination Data Type
- Transformation Logic
- Default Values
- Formatting Rules
- Examples
- Implementation Notes

This document serves as the conceptual foundation for those implementation specifications.

Related Documents

This document should be read together with:

- Synchronization Model
- Field Mapping
- Stage Mapping
- Agent Mapping
- Validation Rules
- Integration Flows Design

Description

TBD